

ICML 2024: You Don't Need 3B Parameters to Build a Generalist Vision Model

DeLVM, from the University of Sydney, Beijing Institute of Technology, and Huawei Noah's Ark Lab, uses data augmentation and knowledge distillation to make a compact autoregressive vision model learn multiple tasks from limited data

Much of what makes language models work is a single, uniform interface: turn everything into tokens and predict the next one. Recent large vision models (LVMs) carry that autoregressive idea into vision, stitching an image and its annotation into a "visual sentence" and generating a segmentation map, a pose skeleton, or a derained image one token at a time. The appeal is obvious: one model, one interface, many tasks.

The catch is cost. The leading autoregressive LVMs run to 3B parameters and beyond, and training them consumes on the order of 400B visual tokens drawn from 1.64B images. That scale is expensive and awkward to deploy on the edge. Worse, the visual world is long-tailed: segmentation has abundant data, while tasks like keypoint detection and deraining are starved. Throw them into one training mixture and the data-rich tasks drown out the data-poor ones, so the model never learns the rare tasks at all.

DeLVM (Data-efficient Large Vision Models), published at ICML 2024, asks a blunt question: do we actually need all that size and data? The paper's answer is no, and the two tools it reaches for are refreshingly ordinary, data augmentation and knowledge distillation. Both are staples elsewhere in vision, yet almost nobody had tried them seriously on autoregressive LVMs. This work fills that gap.

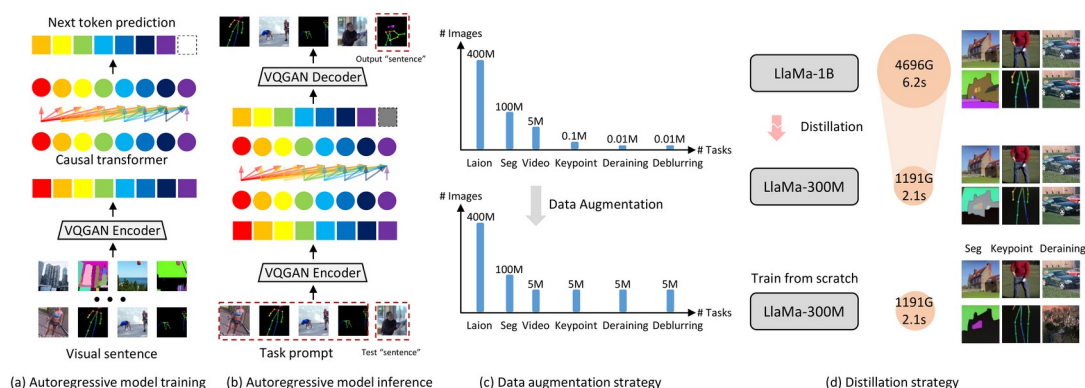


Figure 1. Overview of the DeLVM framework. (a) Training: a VQGAN encoder discretizes each image into a token sequence, and a causal transformer is trained by next-token prediction. (b) Inference: a few input-output image pairs form a task prompt; the model continues it to produce the output for a test image, which the VQGAN decoder turns back into an image. (c) Data augmentation rebalances long-tailed tasks to comparable data volumes. (d) Knowledge distillation uses a LLaMA-1B teacher to guide a compact LLaMA-300M student, which is cheaper and faster at inference.

Paper: <https://arxiv.org/abs/2402.04841> · Code: <https://github.com/ggry/DeLVM>

Why the language-model recipe doesn't transfer

Language models typically train for a single epoch over vast corpora, which is enough to avoid overfitting. Vision rarely has that luxury: many task datasets are small to begin with, so repeating them for many epochs is the norm, and the one-pass schedule simply doesn't carry over. Meanwhile the field keeps scaling models and datasets up. DeLVM goes the other way: instead of collecting more real data, make better use of the limited data on hand, which is exactly what augmentation and distillation are for.

The method keeps the standard autoregressive LVM interface. An image passes through an off-the-shelf VQGAN encoder and becomes 256 discrete tokens; concatenating the tokens of several images yields a visual sentence. A causal transformer is trained with a cross-entropy next-token objective, which is no different in spirit from training a language model. At inference, a handful of image-annotation pairs act as a task prompt; appended with a fresh test image, the model autoregressively generates the next 256 tokens, and the VQGAN decoder reconstructs the result. What the task is gets defined entirely by the examples in the prompt.

Move one: augmentation as free data

The first observation is direct: for a data-starved task, simple random crops and flips buy you almost as much as collecting more real data. Training a LLaMA-300M for a fixed 12K steps on subsets of SA-1B and on augmented COCO-Pose, the authors compare adding real data against adding augmented data.

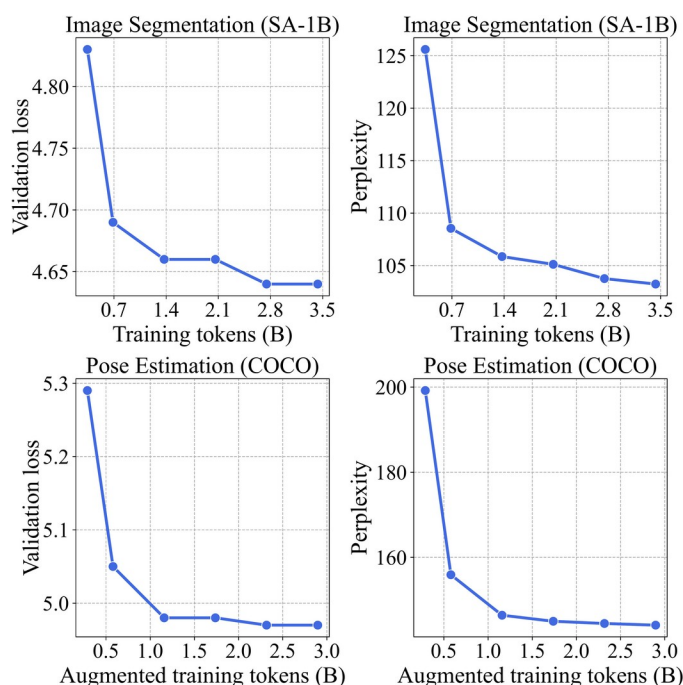


Figure 2. Augmentation matches the effect of introducing new data. Top row, the data-rich segmentation task (SA-1B): validation loss and perplexity keep dropping as training tokens grow. Bottom row, the data-poor pose-estimation task (COCO-Pose, originally about 0.03B tokens): scaling the data through augmentation alone reproduces almost the same downward trend as real new data.

The two rows nearly coincide. For the data-rich segmentation task, more real data steadily lowers validation loss and perplexity; for the data-poor pose task, padding the data out with augmentation traces almost the same curve. Augmentation, in other words, stands in as cheap new data. That matters, because it reframes the long-tail problem from "where do we find more data" to "how do we top up what we already have."

Once a single task is topped up, the real test is training several tasks together. The authors compare three ways of handling the imbalance: training directly on the raw unbalanced data; re-sampling the rare tasks, that is, repeating the minority tasks' examples until the counts even out; and using augmentation to expand every task to a comparable data volume. The differences show up directly in what the model generates on the same set of test inputs, which the figure below lays side by side across segmentation, pose, and deraining.

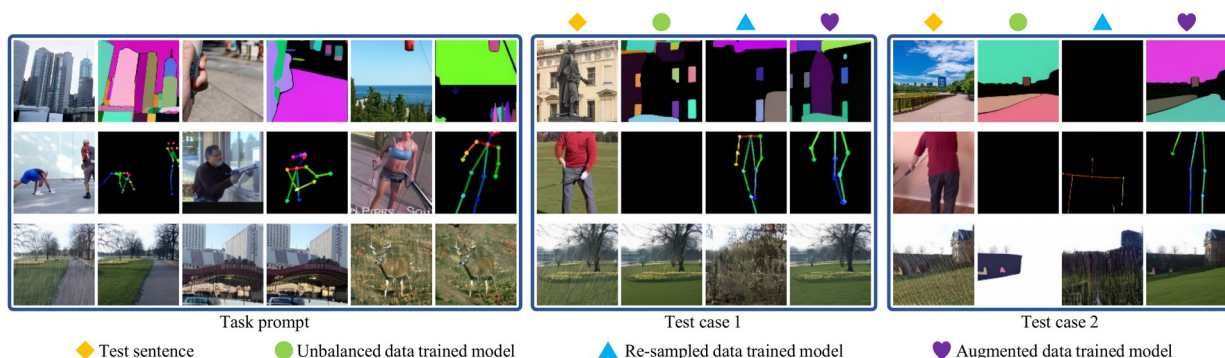


Figure 3. Inference results under different balancing schemes. The diamond marks the test input. The model trained on unbalanced data (circle) is lopsided, doing well only on segmentation; re-sampling (triangle) makes things worse overall; rebalancing with augmentation (heart) yields cleaner, more sensible outputs across segmentation, pose, and deraining.

The takeaway is clean. The raw mixture leaves the model lopsided, strong only on the data-heavy segmentation task. Naive re-sampling doesn't fix it and actually collapses on the scarce tasks. Only balancing the three tasks to comparable data volumes with augmentation gives the best overall performance across all three at once.

Move two: distillation to shrink the model

The second lever is knowledge distillation, which the authors describe as the first use of distillation on autoregressive LVMs. The recipe itself is standard: a larger LLaMA-1B teacher guides a compact LLaMA-300M student, so that during training the student not only fits the ground-truth annotation but also aligns to the teacher's output distribution. The real question is the setting: in language distillation the teacher hands over soft probabilities over a vocabulary, whereas here the model generates images token by token, and both teacher and student emit long strings of discrete visual tokens. Under that generative setting, can distillation still transfer the teacher's ability to the student? Single-task results, shown in the figure below, give a fairly direct answer to that question.

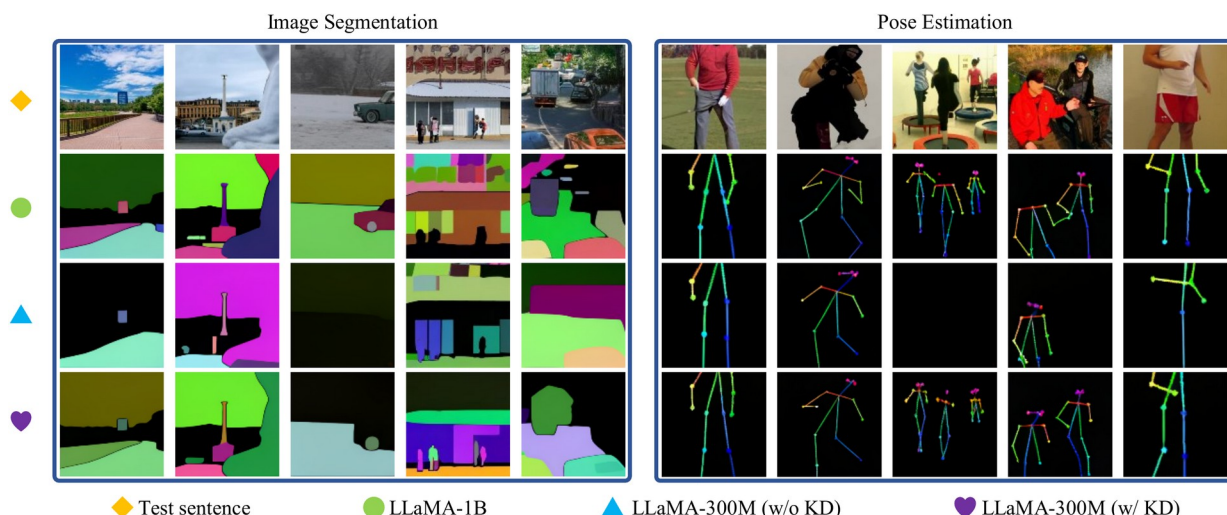


Figure 4. Single-task inference results. The diamond is the test input, the circle is the LLaMA-1B teacher, the triangle is a LLaMA-300M trained from scratch, and the heart is a LLaMA-300M trained with knowledge distillation. The teacher is best, and the distilled student (heart) is clearly closer to it, on both segmentation and pose estimation, than the same-size model trained from scratch (triangle).

It does help. In both single-task and multi-task settings, distillation consistently improves the compact student on segmentation, pose estimation, and deraining. Visually, the distilled LLaMA-300M produces segmentation maps and pose skeletons that sit closer to the LLaMA-1B teacher, while the from-scratch model of the same size looks noticeably rougher. On the Pascal-5i foreground-segmentation benchmark downstream, the distilled-and-fine-tuned LLaMA-300M also clearly beats its from-scratch counterpart.

An easy-to-miss detail: multi-task training must be shuffled

One of the paper's most instructive ablations concerns the order of the task data. Train the tasks sequentially rather than shuffled, one after another (segmentation on SA-1B, then pose on COCO-Pose, then deraining on Rain13K), and the model suffers catastrophic forgetting: it performs well only on the task it saw most recently and is nearly useless on the earlier ones. The authors report in a table that perplexity on the early tasks climbs from a normal hundred-something all the way into the thousands. The figure below places the inference results from three training stages together, making the forgetting easy to see.

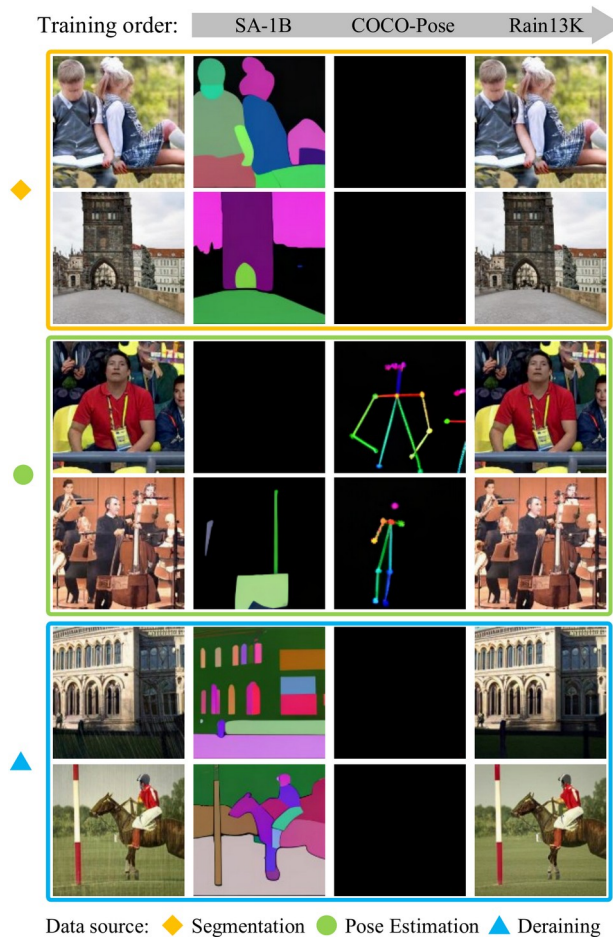


Figure 6. Inference results at different stages of sequential (unshuffled) training. The three rows correspond to test samples from segmentation, pose, and deraining data sources. Regardless of which task the prompt points to, the model's competence collapses onto whatever task it trained on last, the signature of catastrophic forgetting, showing that multi-task LVM training has to be shuffled.

The lesson is that packing several tasks into one autoregressive LVM makes how you organize the data as important as the data itself. The authors also note that the prompt's background color propagates into the output: black-background prompts make a simple grayscale-threshold post-processing step a reliable way to extract the foreground mask. These small engineering details are exactly what makes a compact model actually work across tasks.

Both moves together: even an 80M model holds up

Stacking augmentation and distillation, the authors push to a more extreme case: a LLaMA-80M with just 80M parameters, which the paper reports reaches 83.04% top-1 accuracy on ImageNet classification under this recipe. For a compact autoregressive model trained with a generative objective, that number is genuinely surprising, and it hints that generation and understanding might be learnable together in one model. The table below summarizes two representative results.

Table 1. Two representative DeLVM results (numbers from the paper).

Setting	Metric	Result
LLaMA-300M (distilled + fine-tuned)	Pascal-5i foreground-seg mIoU (split 0/1/2/3)	18.58 / 21.32 / 19.90 / 21.08
LLaMA-80M (augmentation + distillation)	ImageNet top-1 accuracy	83.04%
SA-1B data 1% → 10%	Validation loss / perplexity drop:	0.19 / 22.4

What it means, and where it stops

DeLVM's message is clean: building a capable autoregressive vision model need not take 3B parameters and hundreds of billions of tokens. Two classical techniques, augmentation to rebalance long-tailed tasks and distillation to compress a large teacher, let a compact model trained on limited data perform well across segmentation, pose estimation, and deraining. An 80M version even touches 83.04% on ImageNet, pointing toward cheaper, more deployable generalist vision models.

The boundaries are worth stating too. The evaluation centers on a handful of tasks, segmentation, pose estimation, and deraining, so the task coverage is still narrow; the model leans on an off-the-shelf VQGAN tokenizer, whose reconstruction ceiling caps the quality of fine-grained outputs; and the eye-catching 83.04% comes from a specific setup whose reach to broader understanding tasks needs more validation. Still, as a study in spending less by using old tricks correctly, DeLVM offers a practical direction: before scaling the model up again, check whether you've wrung all you can out of the data you already have.